

Table 4
 Relationship between external HTC and surface roughness for copper tubes.

Boiling mode	Saturation temperature [°C]	Dependence of external HTC on surface roughness
Pool boiling	5	$h_o \propto R_a^{0.110}$
Pool boiling	25	$h_o \propto R_a^{0.125}$
Falling film	5	$h_o \propto R_a^{0.167}$
Falling film	25	$h_o \propto R_a^{0.206}$

7. Falling film results

7.1. Influence of roughness and saturation temperature on heat transfer

The HTCs as a function of heat flux for falling film boiling conditions are shown in [Fig. 15](#). A range of different roughness copper tubes were tested at saturation temperatures of 5 °C and 25 °C with a film Reynolds number of approximately 2100.